MLTL — PRACTICAL 08

Transfer Learning with Images

Gayatri Vidhate · CM23039 · MobileNet Feature Extraction · Custom Image Classes

MobileNet ready — 3 classes · 0 images

STEP 1 — ADD CLASSES & UPLOAD IMAGES

1

Apple 🍎 — 0 images — + Add Images

Banana 🍌 — 0 images — + Add Images

Orange 🍊 — 0 images — + Add Images

+ Add New Class

STEP 2 — TRAIN CLASSIFIER

2

EPOCHS	BATCH SIZE
40	8
LEARNING RATE	**HIDDEN UNITS**
0.001	64

Train Model

Gayatri Vidhate · CM23039 · MobileNet Feature Extraction · Custom Image Classes

MobileNet ready — 3 classes · 16 images

STEP 1 — ADD CLASSES & UPLOAD IMAGES

1

Apple 🍎 — 5 images — + Add Images

Banana 🍌 — 6 images — + Add Images

Orange 🍊 — 5 images — + Add Images

+ Add New Class

STEP 2 — TRAIN CLASSIFIER

2

EPOCHS	BATCH SIZE
40	8
LEARNING RATE	**HIDDEN UNITS**
0.001	64

Train Model

ACTIVITY LOG

Clear

5 images | 6 images | 5 images

+ Add Images | + Add Images | + Add Images

+ Add New Class

STEP 2 — TRAIN CLASSIFIER

2

EPOCHS: 40

BATCH SIZE: 8

LEARNING RATE: 0.001

HIDDEN UNITS: 64

Train Model

Epoch 40 / 40 — 100%

0.9234	69%	40
LOSS	ACCURACY	EPOCH

STEP 3 — EVALUATION

3

ACCURACY PER CLASS

CLASS	ACCURACY	SAMPLES
Apple 🍎	60%	5
Banana 🍌	83%	6
Orange 🍊	100%	5

CONFUSION MATRIX

Actual \ Predicted	Apple 🍎	Banana 🍌	Orange 🍊
Apple 🍎	3	0	2
Banana 🍌	0	5	1
Orange 🍊	0	0	5

STEP 4 — PREDICT ON NEW IMAGE

4

CONFIDENCE PER CLASS

Orange 🍊 100% 5

Orange 🍊 0 0 5

STEP 4 — PREDICT ON NEW IMAGE

4

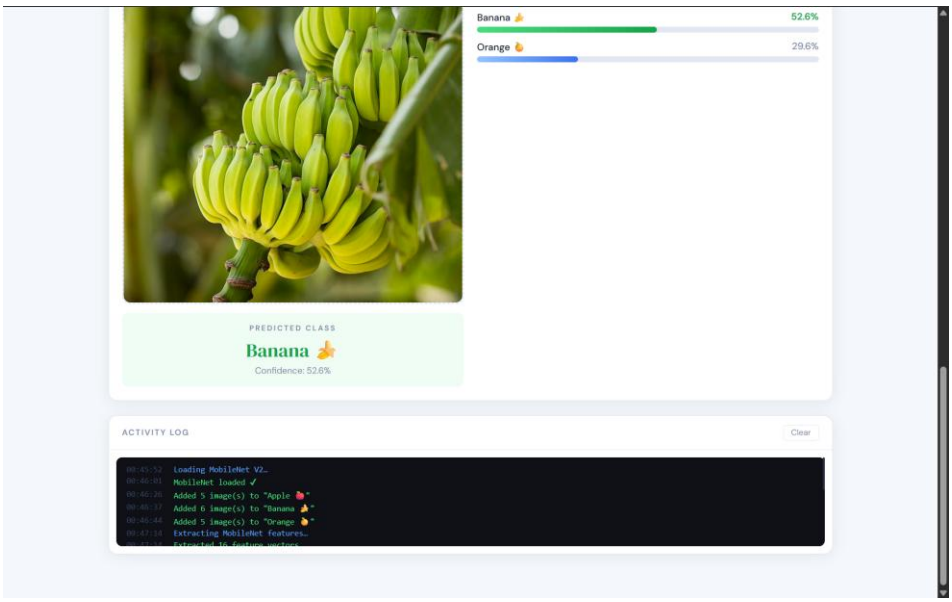
PREDICTED CLASS

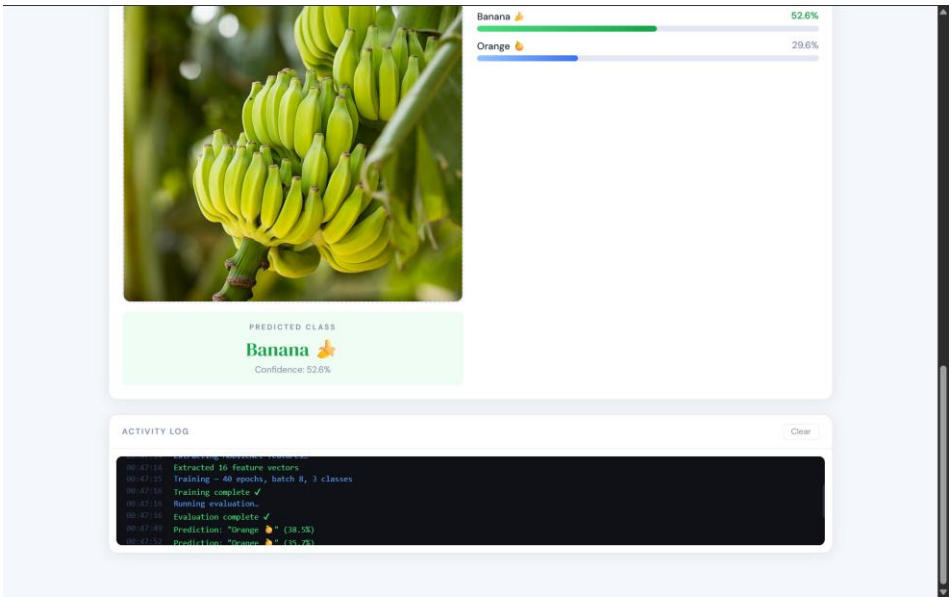
Banana 🍌

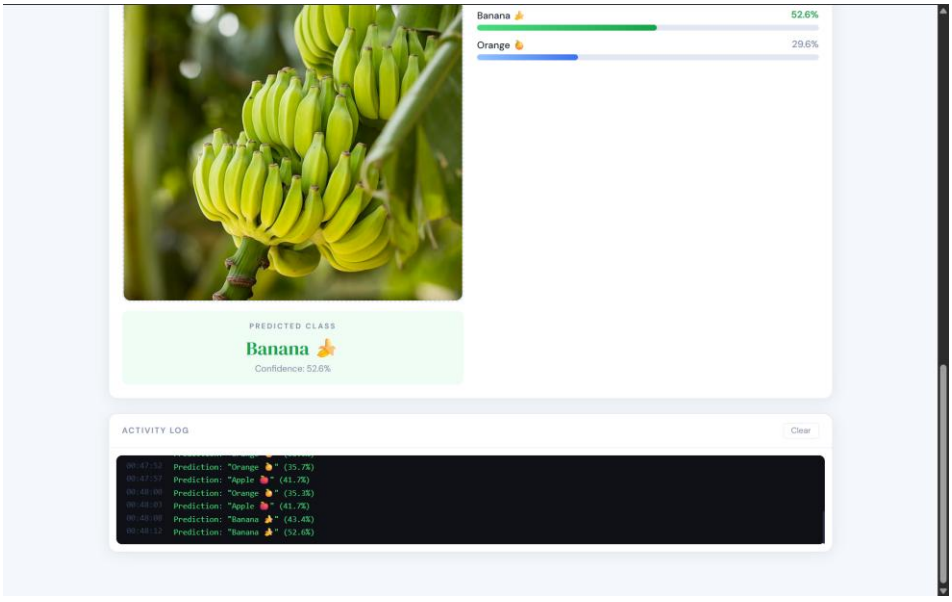
Confidence: 52.6%

CONFIDENCE PER CLASS

Apple 🍎	17.9%
Banana 🍌	52.6%
Orange 🍊	29.6%


Banana 🍌 52.6%
Orange 🍊 29.6%
PREDICTED CLASS
Banana 🍌
Confidence: 52.6%
ACTIVITY LOG
Clear
00:45:52 Loading MobileNet V2…
00:46:01 MobileNet loaded ✓
00:46:26 Added 5 image(s) to "Apple 🍎"
00:46:37 Added 6 image(s) to "Banana 🍌"
00:46:44 Added 5 image(s) to "Orange 🍊"
00:47:14 Extracting MobileNet features…


Banana 🍌 52.6%
Orange 🍊 29.6%
PREDICTED CLASS
Banana 🍌
Confidence: 52.6%
ACTIVITY LOG
Clear
00:47:14 Extracted 16 feature vectors
00:47:15 Training — 40 epochs, batch 8, 3 classes
00:47:16 Training complete ✓
00:47:16 Running evaluation…
00:47:16 Evaluation complete ✓
00:47:49 Prediction: "Orange 🍊" (38.5%)


Banana 🍌 52.6%
Orange 🍊 29.6%
PREDICTED CLASS
Banana 🍌
Confidence: 52.6%
ACTIVITY LOG
Clear
00:47:52 Prediction: "Orange 🍊" (35.7%)
00:47:57 Prediction: "Apple 🍎" (41.7%)
00:48:00 Prediction: "Orange 🍊" (35.3%)
00:48:03 Prediction: "Apple 🍎" (41.7%)
00:48:08 Prediction: "Banana 🍌" (43.4%)
00:48:12 Prediction: "Banana 🍌" (52.6%)